DSC 4107 Homework 1 Solutions

Final answers are rounded to 4 significant figures. Intermediate values are kept unrounded.

Question 1

Let $F = 1000$, $y = 0.08$, $T = 3$.

Q1-1. Annual coupon, $c = 10\%$. The coupon is $C = 0.10 \times 1000 = 100$.

$$P = \sum_{t=1}^{3} \frac{C}{(1+y)^t} + \frac{F}{(1+y)^3} = 100\frac{1 - 1.08^{-3}}{0.08} + \frac{1000}{1.08^3}$$

$$P = 257.7096 + 793.8322 = 1051.54 \approx \mathbf{1052}$$

Calculator: N = 3, I/Y = 8, PMT = 100, FV = 1000 → CPT PV = −1051.54.

Q1-2. Zero coupon. No coupons, so $C = 0$.

$$P = \frac{F}{(1+y)^3} = \frac{1000}{1.08^3} = 793.83 \approx \mathbf{793.8}$$

Calculator: N = 3, I/Y = 8, PMT = 0, FV = 1000 → CPT PV = −793.83.

Q1-3. Quarterly coupon. Per period: $C = 1000 \times 0.10/4 = 25$, $y/m = 0.02$, $n = 12$.

$$P = 25\frac{1 - 1.02^{-12}}{0.02} + \frac{1000}{1.02^{12}} = 264.3835 + 788.4932 = 1052.88 \approx \mathbf{1053}$$

This is higher than Q1-1 by about 1.33 (1052.88 vs 1051.54). Quarterly coupons arrive sooner than annual ones, and with the coupon rate (10%) above the yield (8%), that earlier cash raises the price.

Calculator: N = 12, I/Y = 2, PMT = 25, FV = 1000 → CPT PV = −1052.88.

Q1-4. Continuous compounding, $r = 8\%$. Discount each cash flow by e^{-rt}.

$$P = 100e^{-0.08} + 100e^{-0.16} + 1100e^{-0.24} = 92.3116 + 85.2144 + 865.2908 = 1042.82 \approx \mathbf{1043}$$

Q1-5. Step-up coupon (6%, 8%, 12%). Cash flows are $60, 80, 1120$.

$$P = \frac{60}{1.08} + \frac{80}{1.08^2} + \frac{1120}{1.08^3} = 55.5556 + 68.5871 + 889.0921 = 1013.23 \approx \mathbf{1013}$$

Q1-6. YTM of a 6% bond priced at 950. Solve

$$950 = \frac{60}{1+y} + \frac{60}{(1+y)^2} + \frac{1060}{(1+y)^3}, \quad y = 7.938\% \approx \mathbf{7.938}\%$$

Calculator: N = 3, PV = −950, PMT = 60, FV = 1000 → CPT I/Y = 7.938.

Question 2

Let $F = 100$, coupon rate = 6% paid semi-annually, so $C = 3$. $y = 5\%$, $m = 2$, $y/m = 0.025$, $n = 6$. Discount factor $\mathrm{DF}_k = 1/1.025^k$.

Q2-1. Cash flow table.

Period k	Year	CF	DF	PV of CF
1	0.5	3	0.975610	2.926829
2	1.0	3	0.951814	2.855443
3	1.5	3	0.928599	2.785798
4	2.0	3	0.905951	2.717852
5	2.5	3	0.883854	2.651563
6	3.0	103	0.862297	88.816577
	Total			**102.754063**

The current price is $P_0 = 102.7541$.

Q2-2. Macaulay Duration (weighted average time, in years):

$$D_{\text{Mac}} = \frac{\sum_k (k/m) \times \text{PV}_k}{P_0} = \frac{287.0119}{102.7541} = \mathbf{2.7932} \text{ years}$$

Q2-3. Modified Duration:

$$D_{\text{mod}} = \frac{D_{\text{Mac}}}{1 + y/m} = \frac{2.7932}{1.025} = \mathbf{2.7251}$$

Q2-4. Meaning. Modified duration measures price sensitivity to yield. $D_{\text{mod}} = 2.7251$ means a 1 percentage-point (100 bp) rise in yield lowers the price by about 2.73%. As a first-order approximation, it holds only for small yield changes.

Q2-5. Approximate Modified Duration. Re-price at $y = 5.1\%$ and $y = 4.9\%$:

$$P_+ = 102.4745, \quad P_- = 103.0345, \quad P_0 = 102.7541, \quad \Delta y = 0.001$$

$$\text{Approx. } D_{\text{mod}} = \frac{P_- - P_+}{2P_0 \Delta y} = \frac{0.5600}{0.205508} = \mathbf{2.7251}$$

This matches the exact $D_{\text{mod}} = 2.7251$; a 10 bp move is small, and the linear approximation holds well over that range.

Question 3

Same bond: $F = 100$, $C = 3$, $m = 2$, $y = 5\%$, $P_0 = 102.7541$, $D_{\text{mod}} = 2.7251$.

Q3-1. Exact Convexity.

$$C = \frac{1}{P_0 m^2 (1 + y/m)^2} \sum_{k=1}^{6} \frac{\text{CF}_k}{(1 + y/m)^k} (k^2 + k)$$

The weighted sum $\sum \text{PV}_k(k^2 + k) = 3920.6161$ and the denominator is $P_0 \times 2^2 \times 1.025^2 = 431.8240$, so

$$C = \frac{3920.6161}{431.8240} = \mathbf{9.0792}$$

Q3-2. Finite-difference convexity with $\Delta y = 0.01$ ($y = 6\%$ and 4%):

$$P_+ = 100.0000, \quad P_- = 105.6014$$

$$\text{Approx. Convexity} = \frac{P_- + P_+ - 2P_0}{P_0(\Delta y)^2} = \frac{0.0933}{0.0102754} = \mathbf{9.0805}$$

This is close to the exact 9.0792. A 100 bp step is large enough that the two estimates still differ a little.

Q3-3. Yield rises 200 bp ($\Delta y = +0.02$).

Duration-only model:

$$\frac{\Delta P}{P} \approx -D_{\text{mod}}\Delta y = -(2.7251)(0.02) = -0.05450 = -\mathbf{5.450}\%$$

Duration + convexity model:

$$\frac{\Delta P}{P} \approx -D_{\text{mod}}\Delta y + \frac{1}{2}C(\Delta y)^2 = -0.05450 + \frac{1}{2}(9.0792)(0.02)^2 = -0.05450 + 0.001816 = -\mathbf{5.269}\%$$

Q3-4. Comparison. The duration-only model predicts a loss of 5.450%; adding convexity gives 5.269%. The gap is about 0.18 percentage points. The true loss is $(97.3357 - 102.7541)/102.7541 = -5.273\%$, so the convexity model is much closer.

Duration treats the price-yield relation as a straight line, while the actual curve is convex and bows toward the origin. Over a large yield jump the line overstates the price fall, and the positive convexity term corrects it. That asymmetry favors bondholders: for equal yield moves, the price gain when yields fall is larger than the loss when yields rise.